

Product Basket Analysis

Finding products frequently purchased together using Python & Pandas

Dataset: Online Retail II (UK-based online gift-ware retailer)

Tools: Python, Pandas, SciPy, Matplotlib

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1. Objective & Approach

The goal is an introduction to association-style (market-basket) analysis: find which products customers buy together in the same order, so the business can cross-sell, bundle and place products better. The **order (invoice) ID** is the basket: all items sharing an invoice were bought together.

Method

- Cleaned data: removed cancelled invoices (start with C), adjustments (A), non-positive quantity/price, and non-product codes (POST, DOT, M, BANK CHARGES, etc.).
- Counted each product **once per order** (de-duplicated invoice-product rows).
- Built an order x product matrix and counted co-occurrence of every product pair.
- **Excluded self-pairs** and counted each pair once ($A-B = B-A$) by keeping only $\text{code}(A) < \text{code}(B)$.
- Kept products present in at least 40 orders to avoid noise; ranked pairs by frequency and by lift.

Metrics

- **Pair orders**: number of orders containing both products.
- **Support**: pair orders / total orders. **Confidence ($A \rightarrow B$)**: of orders with A, % that also have B.
- **Lift**: confidence / base popularity of B. Lift > 1 means the products appear together more than chance.

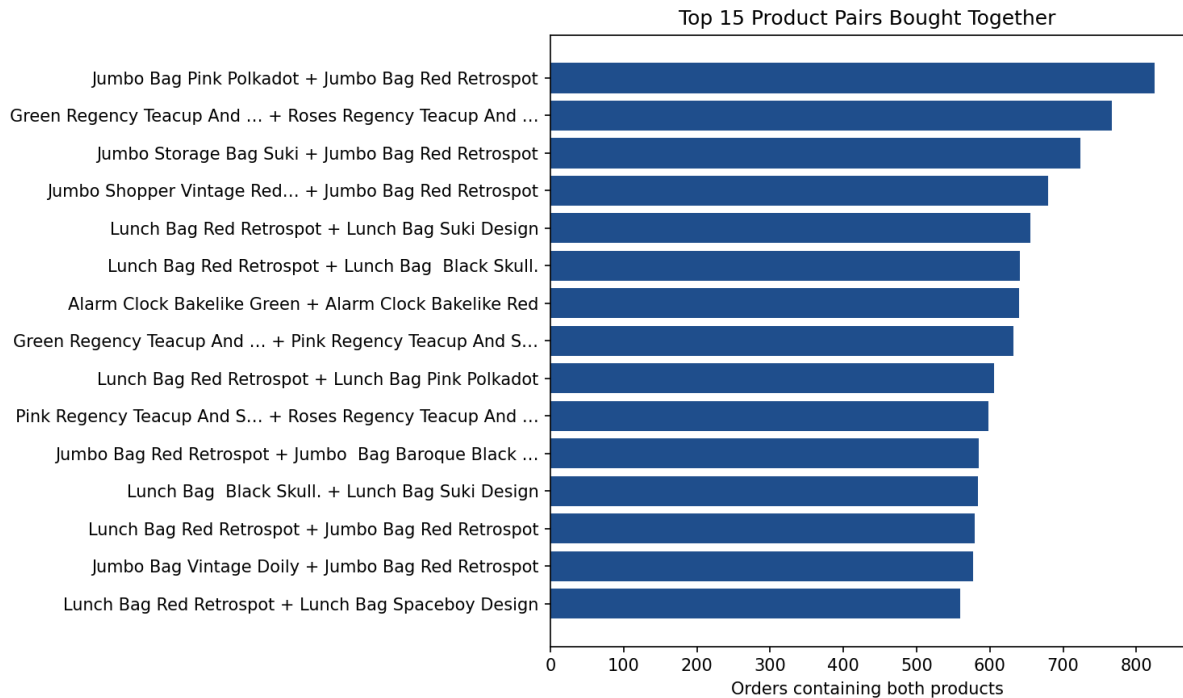
2. Data Overview & Cleaning

Step	Rows / Orders
Raw rows	541,910
After removing cancellations / adjustments	532,619
After removing qty/price <= 0	530,104
After removing non-product codes (final)	527,725
Orders (baskets) / unique products	19,773 / 3,791
Average basket size (unique items)	26.2
Single-item orders (cannot form pairs)	1,500
Orders analysed (2+ items, frequent products)	18,242
Products analysed / pairs evaluated	2,333 / 2,367,021

3. Top Product Pairs

Most frequent pairs (by number of orders containing both items):

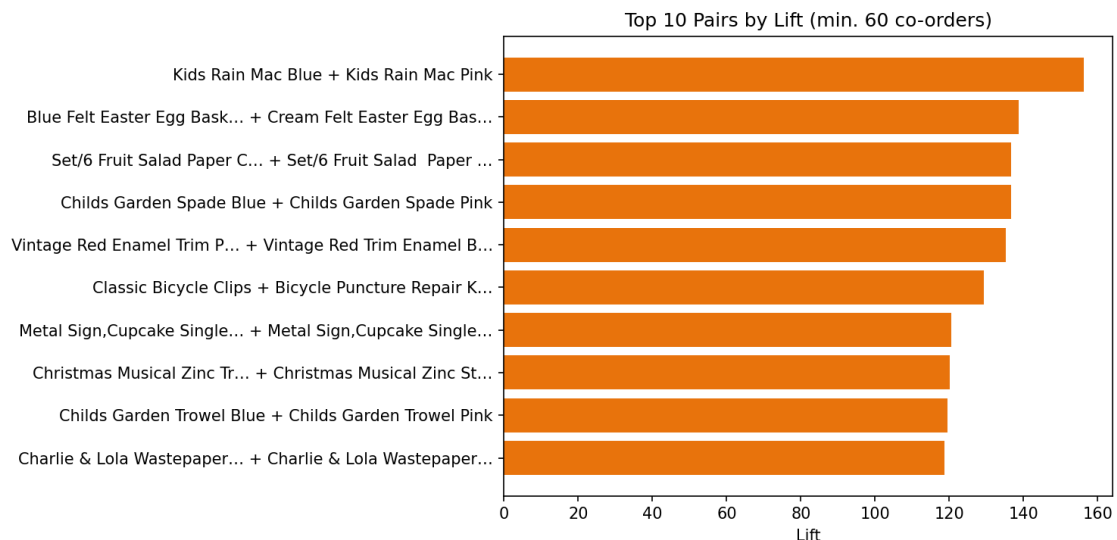
#	Product A	Product B	Orders	Support %	Conf A→B %	Conf B→A %	Lift
1	Jumbo Bag Pink Polkadot	Jumbo Bag Red Retrospot	825	4.523	67.8	39.8	5.97
2	Green Regency Teacup And Saucer	Roses Regency Teacup And Saucer	767	4.205	75.7	72.2	13.01
3	Jumbo Storage Bag Suki	Jumbo Bag Red Retrospot	724	3.969	61.1	34.9	5.38
4	Jumbo Shopper Vintage Red Paisley	Jumbo Bag Red Retrospot	680	3.728	57.9	32.8	5.1
5	Lunch Bag Red Retrospot	Lunch Bag Suki Design	655	3.591	41.9	51.1	5.95
6	Lunch Bag Red Retrospot	Lunch Bag Black Skull.	641	3.514	41.0	50.4	5.88
7	Alarm Clock Bakelike Green	Alarm Clock Bakelike Red	640	3.508	65.6	61.0	11.4
8	Green Regency Teacup And Saucer	Pink Regency Teacup And Saucer	632	3.465	62.4	83.2	14.97
9	Lunch Bag Red Retrospot	Lunch Bag Pink Polkadot	606	3.322	38.7	55.6	6.49
10	Pink Regency Teacup And Saucer	Roses Regency Teacup And Saucer	598	3.278	78.7	56.3	13.52



4. Strongest Associations (by Lift)

Lift shows how much more often two items are bought together than expected by chance (pairs with at least 60 co-orders):

Product A (code)	Product B (code)	Orders	Conf A→B %	Conf B→A %	Lift
Kids Rain Mac Blue (21788)	Kids Rain Mac Pink (21789)	67	78.8	72.8	156.29
Blue Felt Easter Egg Basket (22021)	Cream Felt Easter Egg Basket (85206A)	61	85.9	54.0	138.7
Set/6 Fruit Salad Paper Cups (21088)	Set/6 Fruit Salad Paper Plates (21096)	81	78.6	77.1	136.63
Childs Garden Spade Blue (22514)	Childs Garden Spade Pink (22515)	61	62.9	72.6	136.57
Vintage Red Enamel Trim Plate (23249)	Vintage Red Trim Enamel Bowl (23250)	66	81.5	60.0	135.13
Classic Bicycle Clips (23498)	Bicycle Puncture Repair Kit (23552)	80	80.8	70.2	129.31
Metal Sign,Cupcake Single Hook (82613B)	Metal Sign,Cupcake Single Hook (82613C)	76	66.1	76.0	120.56
Christmas Musical Zinc Tree (22598)	Christmas Musical Zinc Star (22599)	77	81.1	62.6	120.21



5. Key Insights

- **Jumbo bags travel together.** Jumbo Bag Pink Polkadot + Jumbo Bag Red Retrospot is the most frequent pair (825 orders, 4.5% of baskets); 68% of Pink Polkadot buyers also bought Red Retrospot.
- **Regency teacup sets are bought as a collection.** Green, Roses and Pink Regency Teacup & Saucer pair with each other in 600-770 orders each, with lift of 13-15 (13-15x more than chance).
- **Lunch bag designs cluster.** Lunch Bag Red Retrospot pairs with Suki, Black Skull, Pink Polkadot and Spaceboy designs (560-655 orders each): customers buy several designs at once.
- **Colour/variant pairs have the highest lift** (Kids Rain Mac Blue + Pink: lift 156; Child's Garden Spade/Trowel Blue + Pink: lift 120-137). Customers buy matching variants together.
- **Complementary products also appear:** Classic Bicycle Clips + Bicycle Puncture Repair Kit (80 orders, lift 129) and Fruit Salad Paper Cups + Paper Plates (81 orders, lift 137).
- High-lift pairs are small in volume; high-frequency pairs have lower lift but bigger revenue impact. Both views are useful.

6. Recommendations

- **Create bundles / multi-buy offers** for jumbo bag, lunch bag and Regency teacup families (e.g. 'buy 3 designs, save 10%').
- **Add 'Frequently bought together' and 'Complete the set' blocks** on product pages and in the cart using the top pairs table.
- **Cross-sell complementary items:** bicycle clips with repair kit, paper cups with paper plates.
- **Merchandise and pick-pack together:** place high-affinity items side by side (website category pages, catalogue, warehouse picking zones).
- **Stock planning:** forecast and replenish paired items together so one out-of-stock variant does not kill the whole basket.
- **Email campaigns:** after a customer buys one item of a pair, recommend the partner product.

7. Limitations & Next Steps

- Data is from a wholesale-heavy retailer; many orders are bulk buys by trade customers, so patterns may differ for individual shoppers.
- Association is not causation; test recommendations with A/B experiments. Lift on low-volume pairs can be unstable (a minimum order threshold was used).
- Next steps: apply Apriori / FP-Growth (mlxtend) for 3+ item rules, segment by country or customer type, and check seasonality.

8. Interview Questions

What is market-basket analysis? A data-mining technique that finds items frequently bought together in the same transaction, expressed as association rules ($A \rightarrow B$) with support, confidence and lift. It is used for cross-selling, bundling, store layout and recommendations.

Why use order ID? The order ID defines a basket: products with the same order ID were bought together in a single purchase. Using customer ID or date instead would mix separate purchases and create false pairs. Counting each product once per order and excluding self-pairs keeps the counts correct.

Code, data outputs and charts: see the [GitHub repository](#) (basket_analysis.py).